



Science

Stage 9

Paper 1

2022

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	Water is lost from the leaves by transpiration. Water travels upwards through xylem vessels. Water enters the root hair cells by absorption / osmosis.	2	all three correct = 2 marks one or two correct = 1 mark Accept evaporation for transpiration Accept diffusion for osmosis
1(b)	magnesium	1	Accept Mg / Mg ²⁺ but name takes precedence

Question	Answer	Marks	Further Information
2(a)	Si	1	Accept silicon but symbol takes precedence
2(b)	Mg or Ca	1	Accept magnesium or calcium but symbols take precedence Accept Sr / Ba / Ra
2(c)	covalent	1	

Question	Answer	Marks	Further Information
3(a)	B	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
3(b)	increasing the frequency ☐ decreasing the frequency ☐ increasing the amplitude ☒ decreasing the amplitude ☐	1	more than one answer ticked = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence
3(c)	D	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
3(d)	increasing the frequency ☒ decreasing the frequency ☐ increasing the amplitude ☐ decreasing the amplitude ☐	1	more than one answer ticked = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence

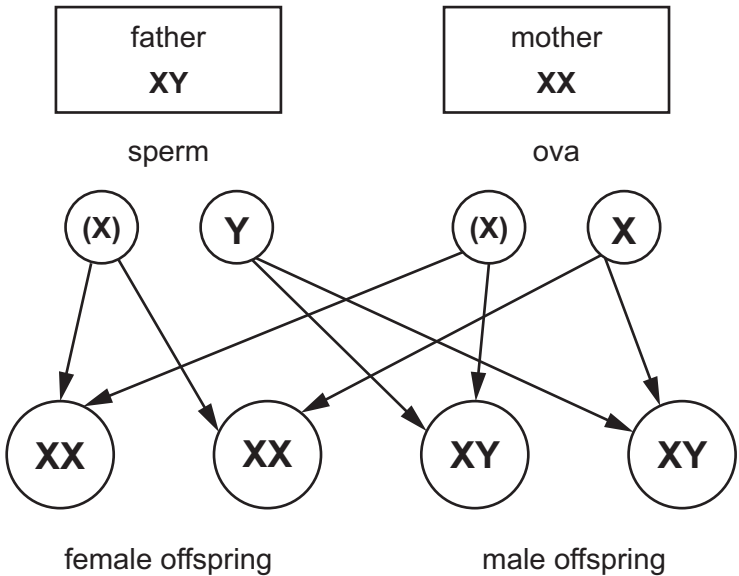
Question	Answer	Marks	Further Information
4(a)	B blue	1	both answers correct for the mark
4(b)(i)	nebula(e)	1	
4(b)(ii)	(idea that) object smashed/collided into (early) Pluto (idea that) ejected particles/mass pulled together by gravity	2	each correct answer = 1 mark

Question	Answer	Marks	Further Information
5(a)	(electric) hot water bath	1	Accept hot water bath heated with Bunsen burner Ignore test-tube holders/tongs/goggles
5(b)	(light) because starch is in part 2 but not in part 1 (and part 4) (chlorophyll) because starch is present in part 2 but not in part 3 (and part 4)	2	each correct answer = 1 mark Accept starch is only in part 2 = 2 marks

Question	Answer	Marks	Further Information
6(a)	density = mass ÷ volume / 316 ÷ 40 7.9 (g / cm ³)	2	correct formula only = 1 mark Note correct answer (with or without working) = 2 marks
6(b)	(idea that) density of hydrogen is (much) lower than the density of iron	1	

Question	Answer	Marks	Further Information
7(a)	convection	1	
7(b)	stops/reduces evaporation / stops steam escaping	1	Accept stops the hot water vapour escaping / stops gas particles escaping / reduces convection above the water Ignore references to heat loss and heat escaping Ignore reference to air escaping but accept gas cannot escape Accept ora, e.g. without a lid steam escapes
7(c)	wood (no mark) because wood is a better insulator / poorer conductor of heat	1	Accept steel (no mark) because less likely to catch fire

Question	Answer	Marks	Further Information
8	any three from the environment changed (idea that) composition of atmosphere changed lack of food lack of light lack of shelter dinosaurs could not escape the danger from the asteroid collision extreme global cooling no/less photosynthesis in plants collision caused a fire that destroyed their habitat	3	each correct answer = 1 mark Accept any suitable suggestion

Question	Answer	Marks	Further Information
9(a)	 The diagram illustrates the inheritance of sex chromosomes. At the top, a box labeled 'father XY' is connected by an arrow labeled 'sperm' to two circles containing '(X)' and 'Y'. To the right, a box labeled 'mother XX' is connected by an arrow labeled 'ova' to two circles containing '(X)' and 'X'. Arrows from the sperm and ova circles point to four circles at the bottom representing offspring. The first two circles are labeled 'XX' and 'female offspring'. The last two circles are labeled 'XY' and 'male offspring'. The connections are: father's (X) to both female offspring; father's Y to both male offspring; mother's (X) to both female and male offspring; and mother's X to both female and male offspring.	2	both sperm and ova line correct = 1 mark offspring line correct = 1 mark Accept YX for male offspring
9(b)(i)	hair colour / ear size / ear position (lop or erect)	1	Accept coat colour / ear shape/length
9(b)(ii)	genetic / genes	1	Accept alleles Ignore DNA / chromosomes

Question	Answer	Marks	Further Information
10(a)	(most reactive) magnesium zinc iron (least reactive) copper	1	all four answers in the correct order for the mark
10(b)	(zinc + copper sulfate) $\longrightarrow$ copper + zinc sulfate	1	both products needed either order for the mark

Question	Answer	Marks	Further Information
11(a)	lamp / bulb	1	
11(b)	(open) switch	1	
11(c)	24 (Ω)	1	
11(d)(i)	cell / battery	1	
11(d)(ii)	(fixed/variable) resistor / bulb / lamp / buzzer / ammeter	1	
11(d)(iii)	current	1	Accept any indication of the correct answer, e.g. circling, ticking or underlining, but answer in the sentence takes precedence

Question	Answer	Marks	Further Information
12(a)(i)	feeding / nutrition	1	
12(a)(ii)	oxygen	1	Accept O ₂ but name takes precedence
12(b)(i)	(carbon dioxide concentration) decreases then increases	2	each correct answer = 1 mark Accept goes down = 1 mark Accept goes up = 1 mark
12(b)(ii)	line showing the concentration is increasing from the existing line to present day	1	Accept straight line or curve Accept any value above 325 p.p.m.
12(c)	any one from sea level change flooding drought extreme weather events	1	Accept ice caps melting Accept named example, e.g. El Niño Accept increased bushfires Accept climate change Accept acid rain

Question	Answer	Marks	Further Information
13	filter (to remove unreacted copper oxide) evaporate (filtrate to half bulk) leave in a warm oven / leave in a warm place / leave to evaporate	3	each correct answer = 1 mark Ignore to crystallise Accept leave on the window ledge Do not accept boil or heat to dryness
Question	Answer	Marks	Further Information
14(a)	1 / one	1	
14(b)	any number between 50 and 10 (°C)	1	
14(c)	more reactive (than lithium, sodium and potassium)	1	

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